

"Kestrel 1.5T: Catalyst efficiency below threshold requires converter and upst

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Area: bulletins

1 Condition

The customer may report an illuminated MIL on the Kestrel 1.5T accompanied by stored DTC P0420 (Catalyst System Efficiency Below Threshold, Bank 1). This code sets when the powertrain control module determines that oxygen storage in the catalyst has degraded and the downstream sensor signal mirrors the upstream signal too closely. Driveability is typically unaffected, but the vehicle will fail an emissions inspection. Refer to the Kestrel Emissions Service Manual :: Catalyst Efficiency Faults for the full code-setting criteria before proceeding.

2 Cause

P0420 on this platform is most often caused by thermal aging or substrate contamination of catalytic converter assembly CAT-5500, which reduces oxygen storage capacity below the PCM threshold. A secondary contributor is a slow or biased upstream oxygen sensor O2U-5100, which skews the fuel trim correction and accelerates apparent catalyst degradation. Confirm the converter as the root cause using the diagnostic flow in Kestrel Emissions Service Manual :: Catalytic Conversion and verify sensor response per Kestrel Emissions Service Manual :: Upstream and Downstream Oxygen Sensors . Do not replace CAT-5500 until exhaust leaks and misfire faults have been ruled out.

3 Repair Procedure

Replace catalytic converter CAT-5500 following PROC-CAT as documented in the Kestrel Emissions Service Manual :: Catalytic Converter Replacement Procedure , then replace upstream oxygen sensor O2U-5100 per PROC-O2-SENSOR in the Kestrel Emissions Service Manual :: Oxygen Sensor Replacement Procedure . Torque the converter flange fasteners and the O2U-5100 sensor to the specifications listed for the Kestrel Emissions Service Manual :: Catalytic Converter (Bank 1 and Bank 2) . After both components are installed, clear P0420, complete a drive cycle, and confirm the catalyst monitor runs to completion. Component overviews are available in the Kestrel Emissions Service Manual if additional reference is needed.